

ibm_marrakesh (real hardware, 156-qubit Heron r2)

4 qubits, 2 layers

ZNE worse on 80% of instances (MAE 3.0× raw)

4 qubits, 5 layers

ZNE worse on 88% of instances (MAE 4.1× raw)

Absolute error $|\langle H \rangle - \langle H \rangle_{\text{ideal}}|$

